Large Language Models for the Economic and Social Sciences:

# Social Media Simulations

Indira Sen, HWS 2025
https://github.com/dess-mannheim/LLM4ESS-HWS25

# Summary of Last Week and Today

## Last session we learned:

- AI for survey development
- In Silico Samples
- Behavioral Experiments
- Validation and Pitfalls

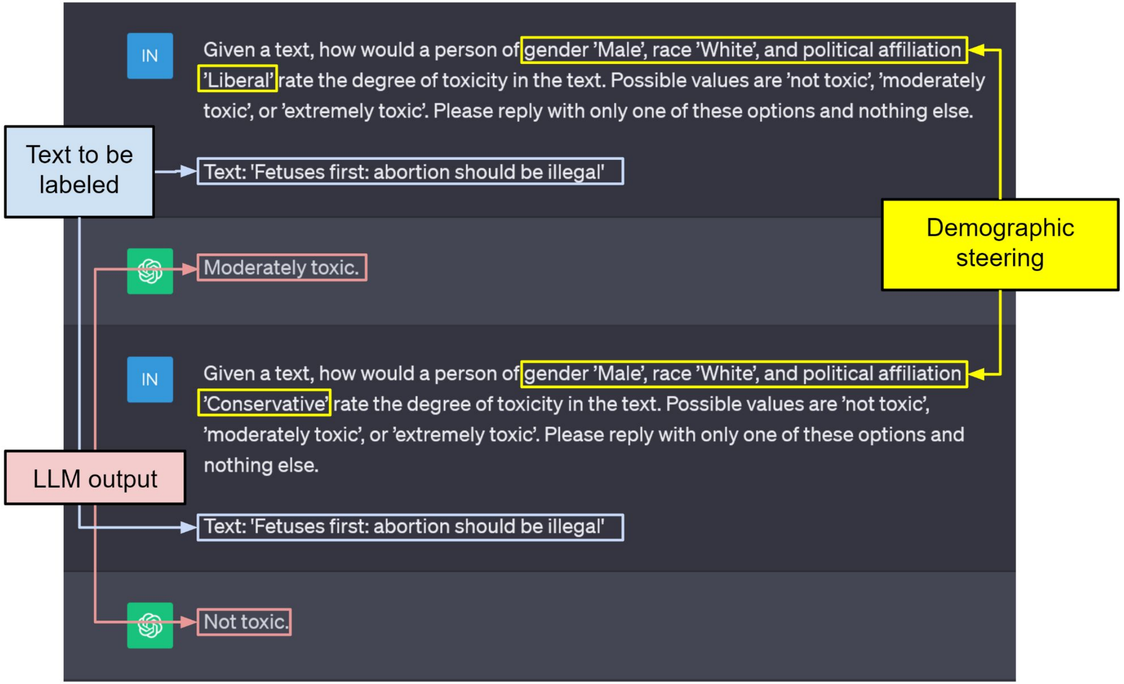
## Today:

- LLM agents
- Traditional Simulations
- Social simulations with LLMs
- Validation and Pitfalls

# Recap: AI-augmented Surveys

# Persona Prompting

- You can add personas to your prompts when doing content analysis to incorporate subjectivity
- Could be demographic personas
  - Age
  - Gender
  - Political leaning
  - …
- Or others — occupation, personality, …



# NLP for Survey Methodology: Silicon Samples

## Out of One, Many: Using Language Models to Simulate Human Samples

**Lisa P. Argyle[1], Ethan C. Busby[1], Nancy Fulda[2], Joshua R. Gubler[1], Christopher Rytting[2] and David Wingate[2]**

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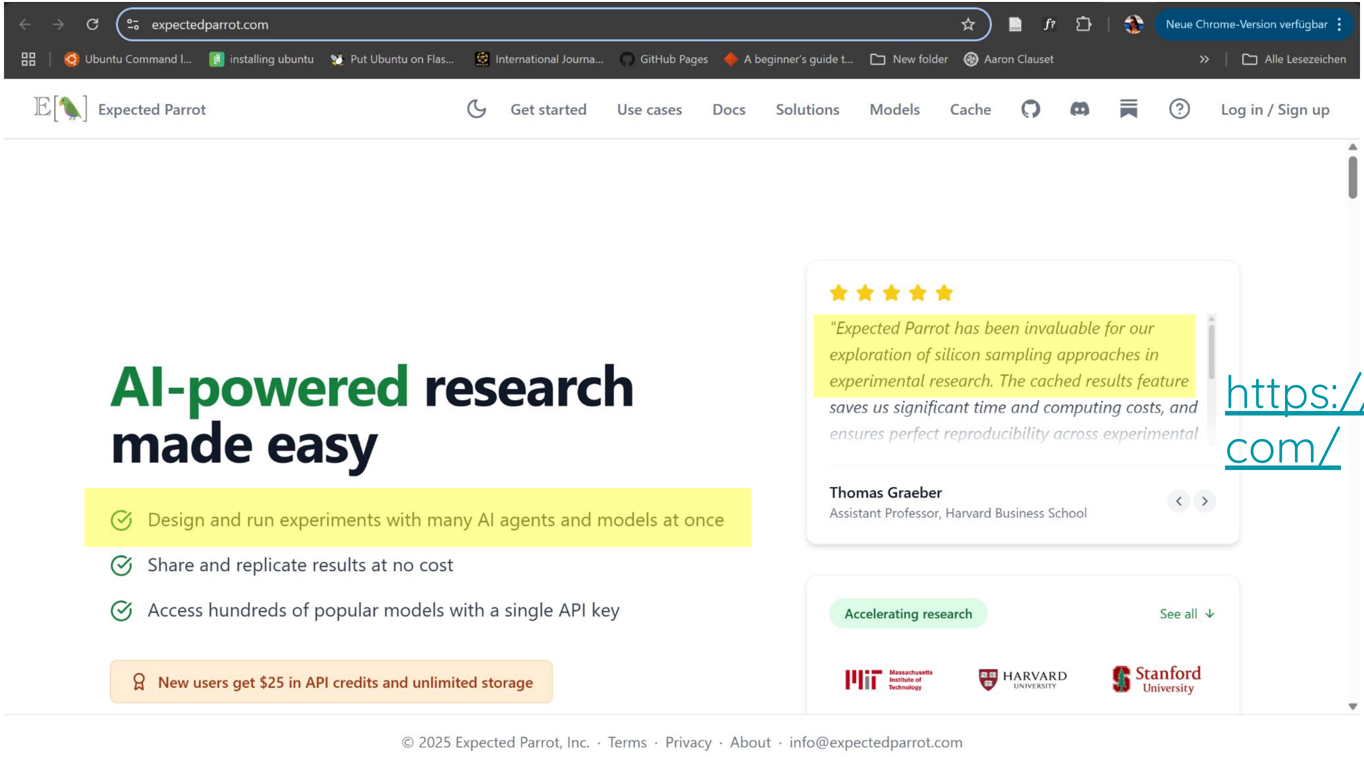
**Abstract**

We propose and explore the possibility that language models can be studied as effective proxies for specific human subpopulations in social science research. Practical and research applications of artificial intelligence tools have sometimes been limited by problematic biases (such as racism or sexism), which are often treated as uniform properties of the models. We show that the "algorithmic bias" within one such tool—the GPT-3 language model—is instead both fine-grained and demographically correlated, meaning that proper conditioning will cause it to accurately emulate response distributions from a wide variety of human subgroups. We term this property *algorithmic fidelity* and explore its extent in GPT-3. We create "silicon samples" by conditioning the model on thousands of sociodemographic backstories from real human participants in multiple large surveys conducted in the United States. We then compare the silicon and human samples to demonstrate that the information contained in GPT-3 goes far beyond surface similarity. It is nuanced, multifaceted, and reflects the complex interplay between ideas, attitudes, and sociocultural context that characterize human attitudes. We suggest that language models with sufficient algorithmic fidelity thus constitute a novel and powerful tool to advance understanding of humans and society across a variety of disciplines.

[Argyle at al., 2023]

# NLP for Survey Methodology: Silicon Samples




https://www.expectedparrot.com/

# Social Simulacra on an individual level

## Out of One, Many: Using Language Models to Simulate Human Samples


Lisa P. Argyle[1], Ethan C. Busby[1], Nancy Fulda[2], Joshua R. Gubler[1], Christopher Rytting[2] and David Wingate[2]

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[Out of One, Many: Using Language Models to Simulate Human Samples]

- Algorithmic bias => algorithmic fidelity => **silicon samples**
- Sociodemographic steering
- algorithmic fidelity: "degree to which the complex patterns of relationships between ideas, attitudes, and sociocultural contexts within a model accurately mirror those within a range of human subpopulations."
- All experiments with GPT3

# Social Simulacra on an individual level

## Out of One, Many: Using Language Models to Simulate Human Samples

**Lisa P. Argyle[1], Ethan C. Busby[1], Nancy Fulda[2], Joshua R. Gubler[1], Christopher Rytting[2] and David Wingate[2]**

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# Silicon Samples: Sociodemographic Steering with Demographic Personas

| | Describing Democrats | Describing Republicans |
|---|---|---|
| Strong Republicans | Ideologically, I describe myself as conservative. Politically, I am a strong Republican. Racially, I am white. I am male. Financially, I am upper-class. In terms of my age, I am young. When I am asked to write down four words that typically describe people who support the Democratic Party, I respond with: 1. **Liberal** 2. **Socialist** 3. **Communist** 4. **Atheist**. | Ideologically, I describe myself as conservative. Politically, I am a strong Republican. Racially, I am white. I am male. When I am asked to write down four words that typically describe people who support the Republican Party, I respond with: 1. **Conservative** 2. **Male** 3. **White (or Caucasian)** 4. **Christian**. |
| Strong Democrats | Ideologically, I describe myself as liberal. Politically, I am a strong Democrat. Racially, I am white. I am female. Financially, I am poor. In terms of my age, I am old. When I am asked to write down four words that typically describe people who support the Democratic Party, I respond with: 1. **Liberal**. 2. **Young**. 3. **Female**. 4. **Poor**. | Ideologically, I describe myself as extremely liberal. Politically, I am a strong Democrat. Racially, I am hispanic. I am male. Financially, I am upper-class. In terms of my age, I am middle-aged. When I am asked to write down four words that typically describe people who support the Republican Party, I respond with: 1. **Ignorant** 2. **Racist** 3. **Misogynist** 4. **Homophobic**. |

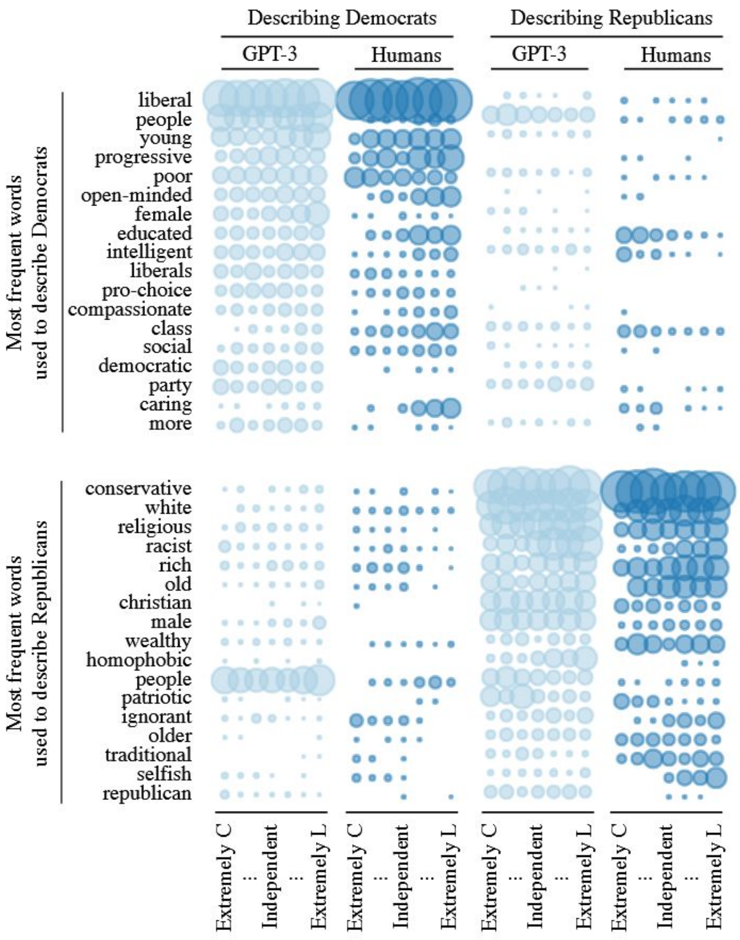
Figure 1: Example contexts and completions from four silicon "individuals" analyzed in Study 1. Plaintext indicates the conditioning context; underlined words show demographics we dynamically inserted into the template; blue words are the four harvested words.

Out of One, Many: Using Language Models to Simulate Human Samples

# Silicon Samples: Answering questions as said demographic persona

- People asked to mention what words they would use to describe Democrats and Republicans

- Humans and LLMs have similar tendencies




[Out of One, Many: Using Language Models to Simulate Human Samples](#)

ding GPT-3 generated words. Bubble
nt the ideology of list writers. GPT-3

# Silicon Samples: Answering closed-form survey questions

- In study 2, Argyle et al., try to see if demographically steered LLMs can predict votes in U.S elections

- Include many variables and measure correlation and proportion of agreement

**Table 1.** Measures of correlation between GPT-3 and ANES probability of voting for the Republican presidential candidate. Tetra refers to tetrachoric correlation. Prop. Agree refers to proportion agreement. GPT-3 vote is a binary version of GPT-3's predicted probability of voting for the Republican candidate, dividing predictions at 0.50.

| Variable | 2012 Tetra. | 2012 Prop. Agree | 2016 Tetra. | 2016 Prop. Agree | 2020 Tetra. | 2020 Prop. Agree |
|---|---|---|---|---|---|---|
| Whole sample | 0.90 | 0.85 | 0.92 | 0.87 | 0.94 | 0.89 |
| Men | 0.90 | 0.85 | 0.93 | 0.88 | 0.95 | 0.88 |
| Women | 0.91 | 0.86 | 0.92 | 0.86 | 0.94 | 0.90 |
| Strong partisans | 0.99 | 0.97 | 1.00 | 0.97 | 1.00 | 0.97 |
| Weak partisans | 0.73 | 0.74 | 0.71 | 0.74 | 0.84 | 0.82 |
| Leaners | 0.90 | 0.85 | 0.93 | 0.87 | 0.95 | 0.89 |
| Independents | 0.31 | 0.59 | 0.41 | 0.62 | 0.02 | 0.53 |
| Conservatives | 0.84 | 0.84 | 0.88 | 0.86 | 0.91 | 0.89 |
| Moderates | 0.65 | 0.77 | 0.76 | 0.78 | 0.71 | 0.77 |
| Liberals | 0.81 | 0.95 | 0.73 | 0.95 | 0.86 | 0.97 |
| Whites | 0.87 | 0.82 | 0.91 | 0.85 | 0.94 | 0.89 |
| Blacks | 0.71 | 0.97 | 0.87 | 0.96 | 0.81 | 0.94 |
| Hispanics | 0.86 | 0.86 | 0.93 | 0.90 | 0.88 | 0.83 |
| Attends church | 0.91 | 0.86 | 0.93 | 0.88 | 0.94 | 0.88 |

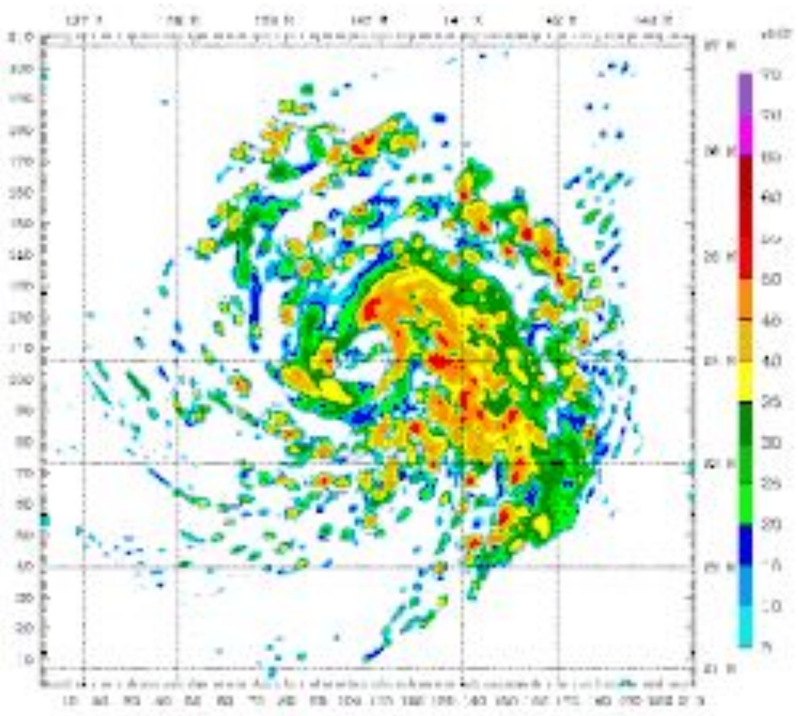
[Out of One, Many: Using Language Models to Simulate Human Samples]

# Traditional Simulations: Agent-Based Models (ABMs)

# Simulations

- A simulation is a program that defines an environment and the behaviors of entities in that environment, then outputs the resulting world

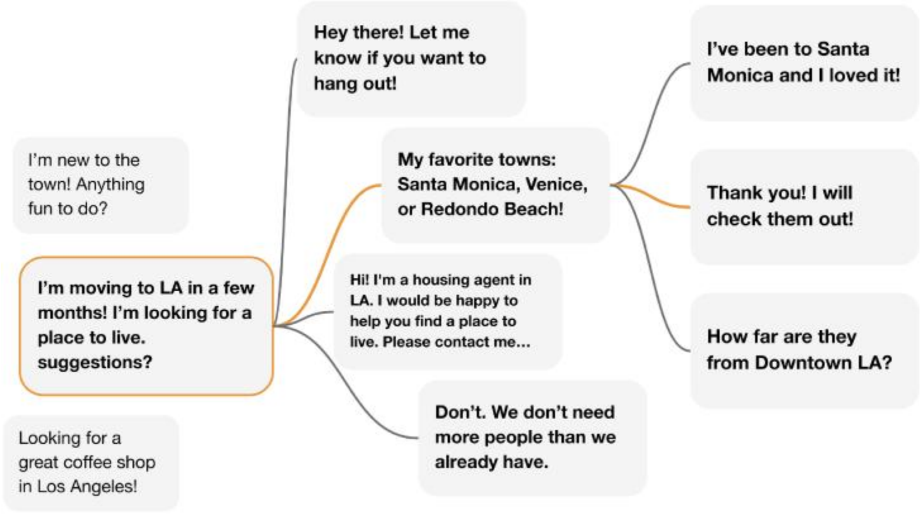
- We see simulations in:
    - Games (The Sims)
    - Movies (The Matrix)
    - Science (Weather simulations)
    - Industry (Flight simulators)



https://en.wikipedia.org/wiki/Computer_simulation

# Why Simulate?

- **Complex systems:** A system composed of many interacting parts in which the emergent outcome of the system is a product of the interactions between the parts and the feedbacks between that emergent outcome and individual decisions
- Micro agents lead to macro outcomes
- Study 'what-if' scenarios or counterfactuals

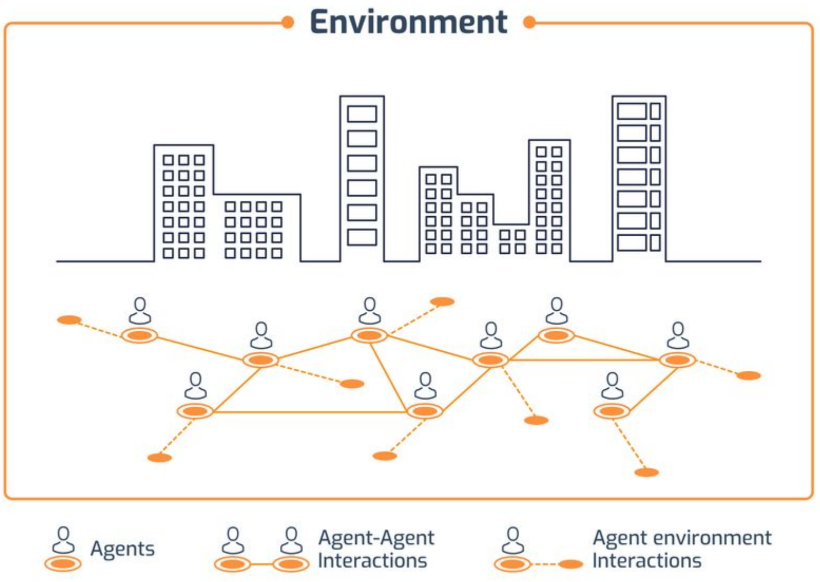



[Social Simulacra: Creating Populated Prototypes for Social Computing Systems]()

# Agent-Based Models (ABM): One Type of Simulation

- Agents: an autonomous individual element with properties and actions in a computer simulation

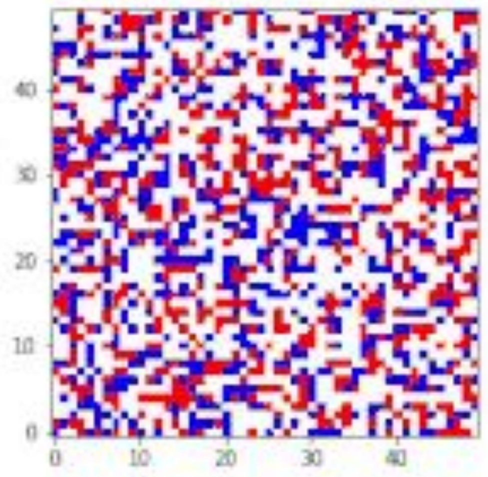
- Agent-Based Modeling (ABM)
  - agents
  - environment
  - description of agent-agent interactions
  - Description of agent-environment interactions




Ramadiah et al., [Agent-Based Simulation of Central Bank Digital Currencies]

# Example: Modeling Segregation

- $N$ X $N$ grid
- Two groups of agents (red and blue)
- only one agent can occupy a space at a time.
- $B_a$ = intolerance of outsiders.
- Even middling values of $B_a$ lead to segregated societies



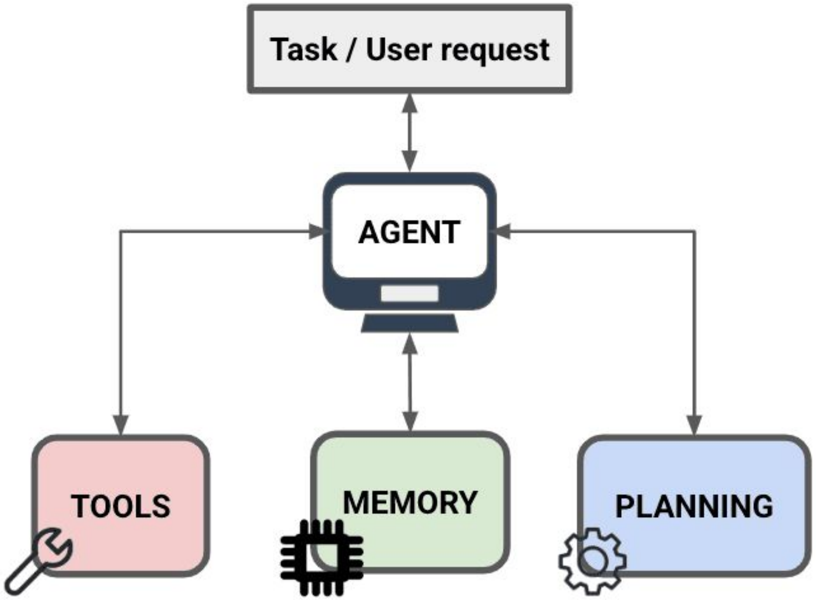

https://en.wikipedia.org/wiki/Schelling%27s_model_of_segregation

# LLM Agents

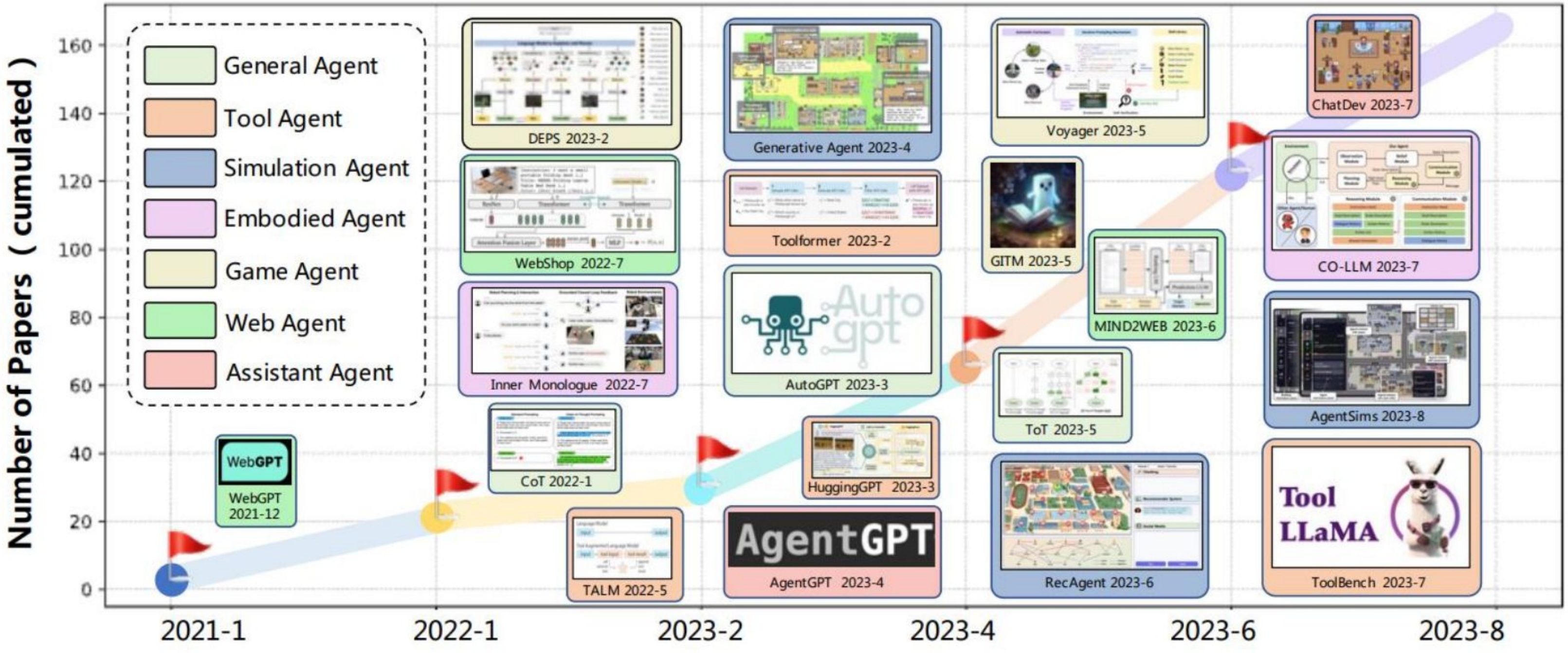
# What are LLM agents and where are they useful?

- LLM agents can autonomously execute complex tasks with **planning** and **memory**

- AI agents already in use for many applications
    - Customer service
    - Health care
    - Education
    - …

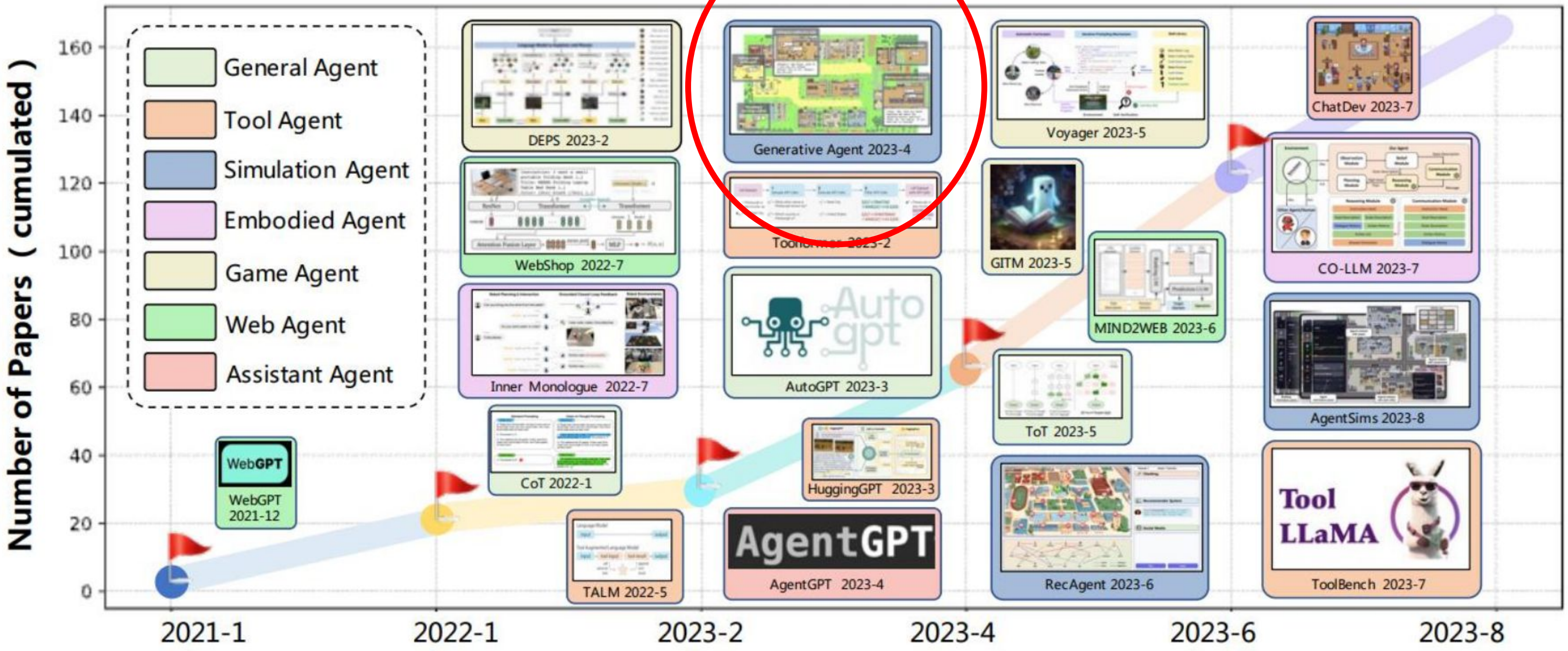


# LLM Agents over time




From Bizer et al.,'s Uni Mannheim course: IE686 Large Language Models and Agents

# LLM Agents over time




From Bizer et al.,'s Uni Mannheim course: [IE686 Large Language Models and Agents]

# Social Simulations with LLMs

# Simulating Social Media Using Large Language Models to Evaluate Alternative News Feed Algorithms

Petter Törnberg[1,c], Diliara Valeeva[2], Justus Uitermark[2], and Christopher Bail[3]

[1]University of Amsterdam, Institute of Language, Logic and Computation (ILLC). [c]p.tornberg@uva.nl. ILLC. P.O. Box 94242, 1090 GE Amsterdam; [2]University of Amsterdam, Amsterdam Institute for Social Science Research (AISSR); [3]Duke University, Departme

[Simulating Social Media Using Large Language Models to Evaluate Alternative News Feed Algorithms]

**Abstract** ia is often criticized for amplifying toxic discourse and discouraging constructive conversations. But designing social media platforms to promote

observers of social media argued these platforms would improve democracy by enabling people to con- ct across social divides (23, 24). Yet most social dia companies were not created to support such h-minded goals (21, 25). For example, Facebook nously evolved from a platform designed to help lege students rate each other's physical attractive- s while Twitter was created to help friends share IS messages with each other in a more efficient nner. None of the world's largest platforms — luding TikTok, Instagram, and Youtube — were signed to promote a constructive public sphere. sessing the impact of these platforms on public nversation may therefore be less productive than

# Social Simulacra: Creating Populated Prototypes for Social Computing Systems

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## ABSTRACT

Social computing prototypes probe the social behaviors that may arise in an envisioned system design. This prototyping practice is currently limited to recruiting small groups of people. Unfortunately, many chal... at a larger scale. ... might behave wh... ustments to the design before the system falls prey to such challenges? We introduce *social simulacra*, a prototyping technique that generates a

[Social Simulacra: Creating Populated Prototypes for Social Computing Systems]

**KEYWORDS**

social computing, prototyping

**ACM Reference Format:**

Joon Sung Park, Lindsay Popowski, Carrie J. Cai, Meredith Ringel Morris, ...rnstein. 2022. Social Simulacra: Creating Populated Prototypes for Social Computing Systems. In *The 35th Annual ACM Symposium on User Interface Software and Technology (To Appear in UIST '22), October 29-November 2, 2022, Bend, OR, USA.* ACM, New York, NY, USA, 18 pages. https://doi.org/XXXXXXX.XXXXXXX

# Emergent behaviors in LLM Agents

[https://youtu.be/iCu8B1Dw9n4?si=tQh-bh2wmmm0BrnA](https://youtu.be/iCu8B1Dw9n4?si=tQh-bh2wmmm0BrnA)

**Generative Agents: Interactive Simulacra of Human Behavior**


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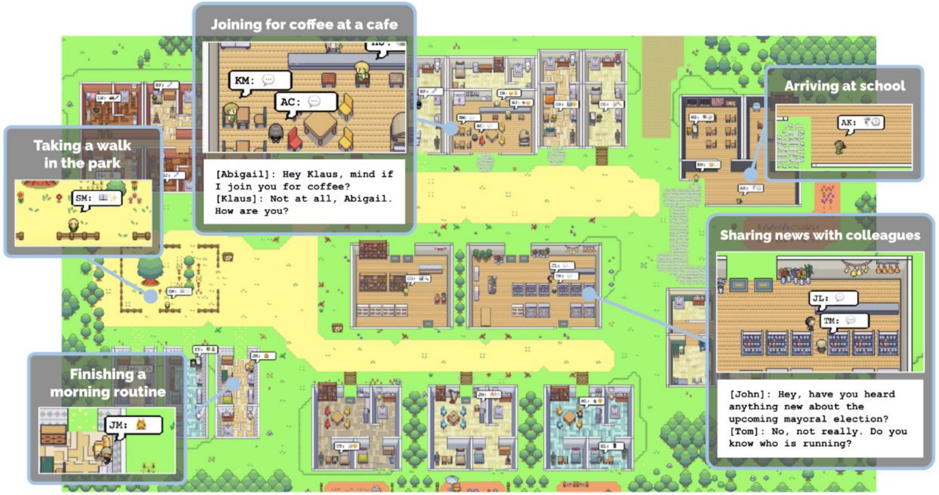


**Figure 1: Generative agents are believable simulacra of human behavior for interactive applications. In this work, we demonstrate generative agents by populating a sandbox environment, reminiscent of The Sims, with twenty-five agents. Users can observe and intervene as agents plan their days, share news, form relationships, and coordinate group activities.**

[Generative Agents: Interactive Simulacra of Human Behavior](#)

# Example simulation:
# Behavior on r/vegan after the introduction of new rules

# Components of an LLM-Simulation

| Input | Process | Output |
| --- | --- | --- |

# Components of an LLM-Simulation

Personas from r/vegan users

Reddit-inspired rules

## Input

1. Agent Population
2. Interaction Rules
   a. Agent–Agent
   b. Agent–Environment
3. Environment
4. LLM(s)

## Process

## Output

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**Personas from r/vegan users**

**Reddit-inspired rules**

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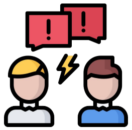
## Process

state
*t = 1*

state
*t = T - 1*

**Discussions after 1 day**



**Discussions after 9 days**

## Output

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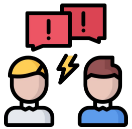
## Process

state
$t = 1$

Discussions after 1 day

state
$t = T - 1$



Discussions after 9 days

## Output

final state
$t = T$

Discussions after 10 days

Estimand or outcome variable

toxicity levels in all discussions

# Types of Social Simulations with LLMs

# Categorizing LLM-Simulations

- Simulation level
- Agent-Agent Interaction
- Agent-Environment interaction

# Categorizing LLM-Simulations

- Simulation level
  - Micro
  - Macro
  - Micro-(meso)-macro
- Agent-Agent Interaction
  - None
  - Constrained
  - Unconstrained
- Agent-Environment interaction
  - No
  - Yes

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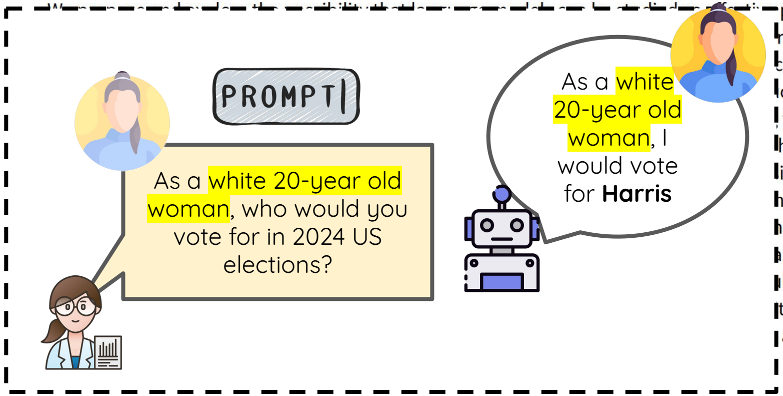
**Abstract**

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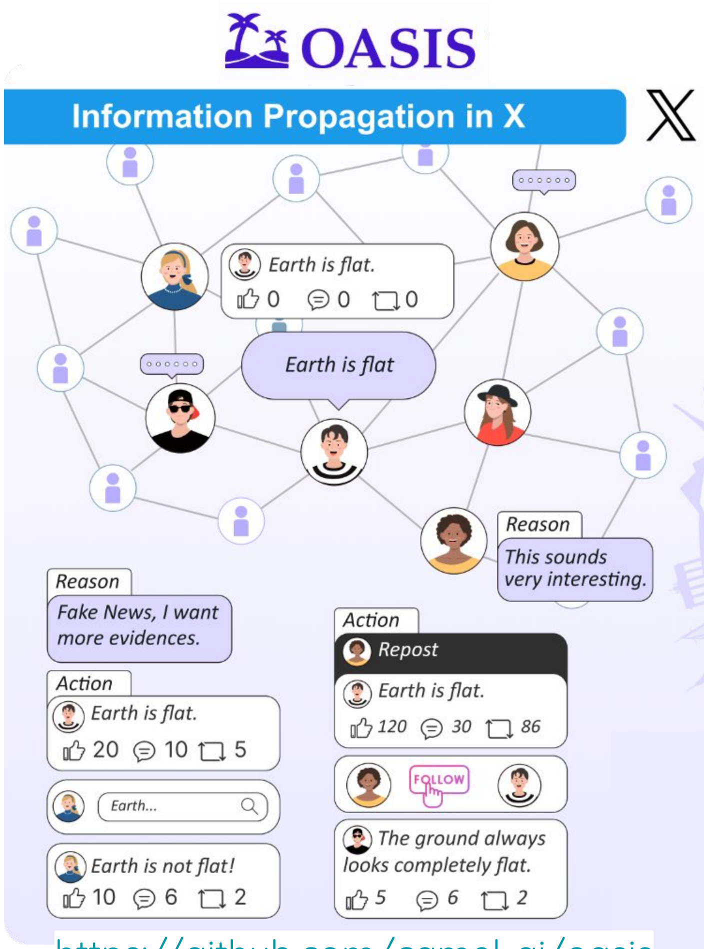


https://github.com/camel-ai/oasis

# Validating Social Simulations

# What could go wrong with simulations?

# Some of the *many* issues...

- Issues with initial simulation parameters [Windrum et al., 2007]
  - Are the rules of interaction realistic?
  - ...
- Issues in modeling
  - How is the final outcome variable measured? [Lundberg et al., 2021]
  - ...
- Issues at different time steps
  - Consistency? [Windrum et al., 2007, Troitzsch, 2014]
  - ...

# What could go wrong with LLM simulations?

# What could go wrong with LLM simulations?

As I look in the mirror, I see my **rich**, **melanin**-infused skin glowing softly. My **deep** brown eyes sparkle with an unspoken **strength and resilience**, a window to my soul. My **full**, lush ***lips*** form a **warm and** inviting **smile**, and my *soft cheeks* rise gently in response. My hair, a riot **of textured coils**, frames my face in a **gravity**-defying halo. It dances to its own beat, wild **and** free, just like me. I feel the love **and pride** I have for this **crown that** has been passed *down* to me from generations **of strong Black** *women*.

GPT4 roleplaying as a black woman asked to **describe herself** (from "Marked Personas: Using Natural Language Prompts to Measure Stereotypes in Language Models" Cheng et al., 2024)

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- Issues with initial simulation parameters [Windrum et al., 2007]
  - Are the rules of interaction realistic?
  - …
- Issues in modeling
  - How is the final outcome variable measured? [Lundberg et al., 2021]
  - …
- Issues at different time steps
  - Consistency? [Windrum et al., 2007, Troitzsch, 2014]
  - …

- Issues with LLM Simulations
  - Simulation bias [Anthis et al. 2025, Larooji and Törnberg 2025]
  - Prompt brittleness [Shu et al., 2024]
  - Sycophancy [Anthis et al. 2025]
  - Hyper Accuracy bias [Aher et al., 2023] / alien intelligence [Kozlowski and Evans 2025]
  - Carbon footprint [Ren et al., 2024]
  - Data Leakage [Barrie andTörnberg, 2025]
  - Temperature fluctuations
  - Guardrails
  - Answer extraction
  - …


A FILM FROM DANIELS
EVERYTHING EVERYWHERE ALL AT ONCE
天馬行空
ORANGE FUZZ


# Validation in the (Quantitative) Social Sciences

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- Measurement theory tells us how social measurements can be evaluated

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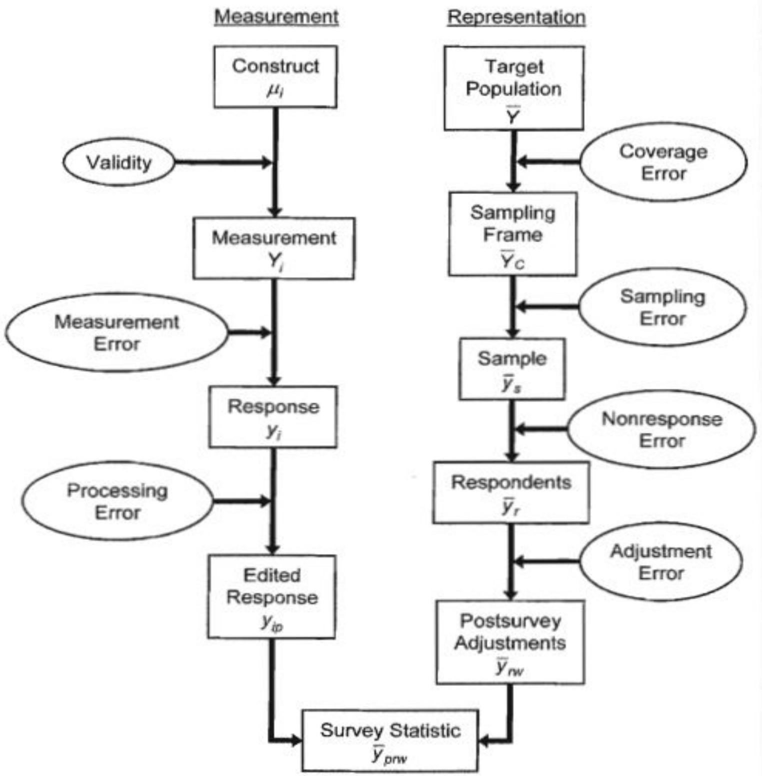
- For survey methodology, we have “error frameworks”, e.g., the Total Survey error framework ([Groves et al., 2008]) that:



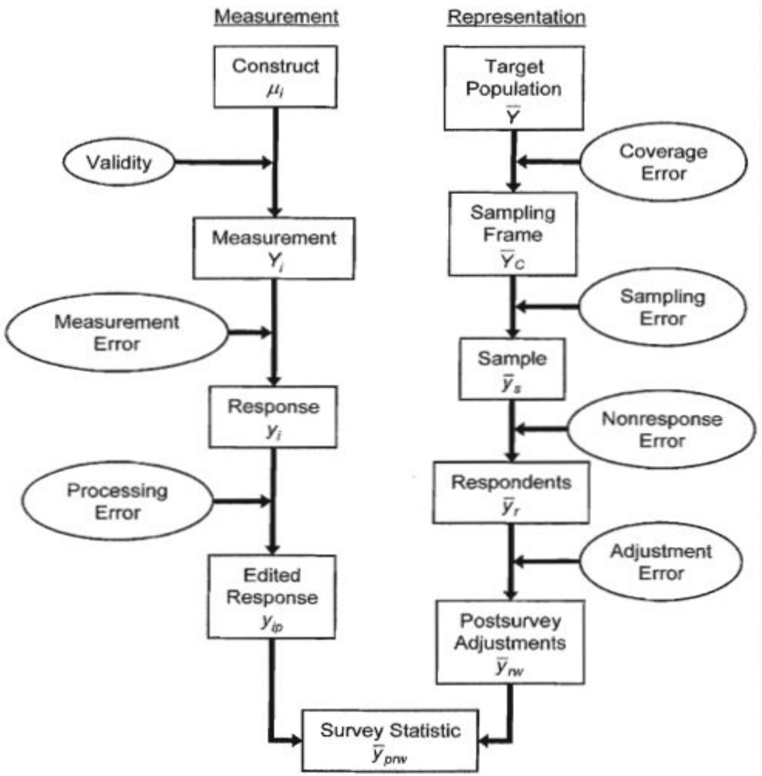
# Validation in the (Quantitative) Social Sciences

- Measurement theory tells us how social measurements can be evaluated

- For survey methodology, we have "error frameworks", e.g., the Total Survey error framework (Groves et al., 2008) that:

  - Systematically identifies and characterizes errors in the process of collecting and building measurement from survey data

  - Helps survey designers understand, reflect on, and document their design choices



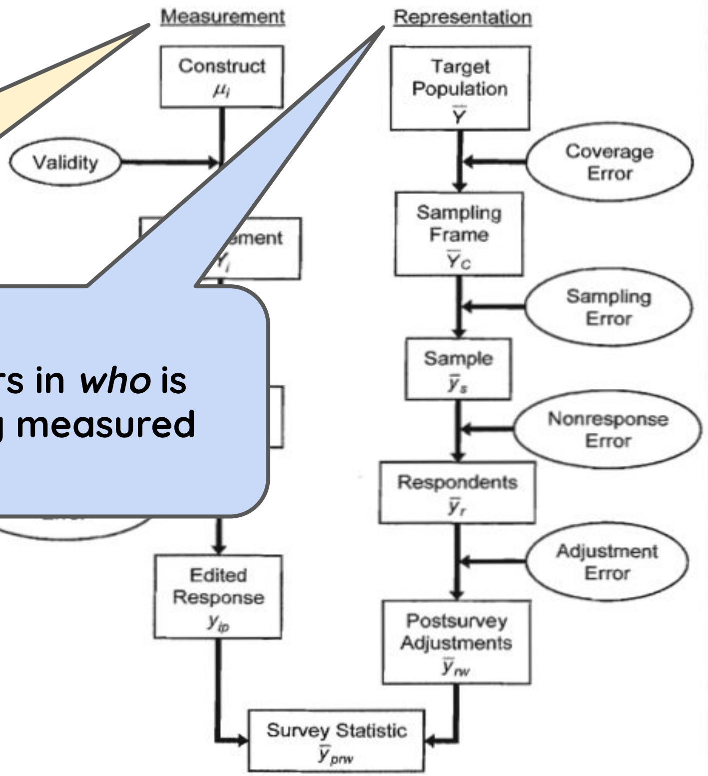
# Validation in the (Quantitative) Social Sciences

- Measurement theory t... social measurements ... evaluated
- For survey methodolo... "error frameworks", e.g., the Total Survey error framework (Groves et al... 2008) that:
    - Systematically identifies and characterizes errors ... collecting and build... from survey data
    - Helps survey desig... reflect on their desi...

**Errors in *how* the construct of interest is defined or operationalized**

**Errors in *who* is being measured**

**Reliability: how *stable* is the measurement?**



Can we apply a similar lense to LLM-simulations?

# Reliability Errors in LLM-simulations

**Persona drift over turns, i.e., LLM reverts to AI persona after turn 5**

## Input

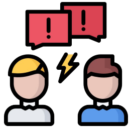
1. Agent Population

2. Interaction Rules
   a. Agent–Agent
   b. Agent–Environment
3. Environment
4. LLM(s)

## Process

state
*t = 1*

state
*t = T - 1*



## Output

final state
*t = T*

Estimand or outcome variable

# Measurement Errors in LLM-simulations

## Input

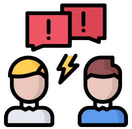
1. Agent Population

2. Interaction Rules
   a. Agent–Agent
   b. Agent–Environment
3. Environment
4. LLM(s)

## Process

state
$t = 1$

state
$t = T - 1$



**LLM guardrails prevent agents from using slurs**

## Output

final state
$t = T$

Estimand or outcome variable

# Representation Errors in LLM-simulations

## Input

1. Agent Population

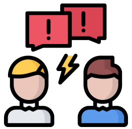
2. Interaction Rules
   a. Agent–Agent
   b. Agent–Environment
3. Environment
4. LLM(s)

LLMs cannot simulate low-income vegans

## Process

state
*t = 1*

state
*t = T - 1*



## Output

final state
*t = T*

Estimand or outcome variable

# Mitigating Issues in Social Simulations with LLMs

Table 1: LLM social simulations must address five key challenges.

| Challenge | Description | Promising Directions |
|---|---|---|
| Diversity | Generic and stereotypical outputs that lack human diversity | Inject humanlike variation in training, tuning, or inference (e.g., interview-based prompting, steering vectors) |
| Bias | Systematic inaccuracies when simulating particular human groups | Prompt with implicit demographic information; minimize accuracy-decreasing biases rather than all social biases |
| Sycophancy | Inaccuracies due to excessively user-pleasing outputs | Reduce the influence of instruction-tuning; instruct LLM to predict as an expert rather than roleplay a persona |
| Alienness | Superficially accurate results generated by non-humanlike mechanisms | Simulate latent features; iteratively conceptualize and evaluate; reassess as mechanistic interpretability advances |
| Generalization | Inaccuracies in out-of-distribution contexts, limiting scientific discovery | Simulate latent features; iteratively conceptualize and evaluate; reassess as generalization capabilities advance |

[Position: LLM Social Simulations Are a Promising Research Method]

# Discussion

- Make a group of 3-4 people
- Think of something that you’d like to simulate with LLMs:
  - Doesn’t have to be about social media
  - Think of the different types of LLM simulations you saw today
- Discuss:
  - Whether its a macro or micro simulation
  - The type of agents
  - How you would validate the simulation
  - The goal of the simulation — would you make policies based on this simulation?

# Summary, Resources, and Next Week

- This session we learned:
    - Applications of LLMs to survey methodology
    - In Silico samples: Replacing survey respondents with LLMs
    - Improving In Silico samples
- Other pointers for LLM Social Simulations:
    - Petter Törnberg's talk on Social Media Simulations with LLMs: https://www.youtube.com/watch?v=BnWRfi0JqzE
    - IE 686: Large Language Models and Agents, Uni Mannheim
    - CS 222: AI Agents and Simulations,Stanford University
- Next week: Project Consultations

Questions?